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Offset hole for TPLO compression

Abstract

TPLO plate includes first and second distal holes and a body with a proximal portion positioned over a proximal tibia segment and a distal portion extending along a longitudinal axis and positioned over a distal tibia segment. The first hole includes an elongated portion extending along the longitudinal axis and an offset portion extending distally from the elongated portion along an offset axis at an angle relative to the longitudinal axis. The second hole extends through the distal portion along the longitudinal axis in alignment with the elongated portion and includes a sloped compression surface along a distal portion. The first hole is such that, when a distal compression to osteotomy cut is applied via insertion of a bone fixation element into the distal portion, the element translates along the offset axis from a central axis to an intersection point to provide a cranial compression to the cut.

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Background/Summary

BACKGROUND

(1) A Tibial Plateau Leveling Osteotomy (TPLO) is a surgical procedure for stabilizing a canine stifle joint, which is comparable to a human knee joint, after a ruptured cranial cruciate ligament (CCL). When the CCL is ruptured or torn, the animal's tibia slides forward with respect to its femur, making it difficult to walk and causing pain. In order to stabilize the joint, a curvilinear cut is made to the upper portion of the tibia. This cut portion of the tibia is then rotated to create a more level plane or surface on the top of the tibia, on which the femur can rest. The cut and repositioned portion of the tibia is then secured to the lower portion of the tibia using a TPLO plate.

(2) TPLO plates are generally sized and shaped to extend along the two portions of the tibia to facilitate healing of the tibia in its new configuration. In some cases, however, where the cut portion is not properly seated on the lower portion of the tibia or where the osteotomy cut is not sufficiently compressed against the lower portion of the tibia, the bone may fail to heal properly.

SUMMARY OF THE INVENTION

(3) The present disclosure relates to a Tibial Plateau Leveling Osteotomy (TPLO) plate for providing a cranial and a distal compression of an osteotomy cut. The TPLO plates includes a body extending longitudinally from a proximal end to a distal end and defined via a first surface which, in an operative configuration, faces away from a bone and a second surface which, in the operative

configuration, faces toward the bone, the body including a proximal portion configured to be positioned over a cut and repositioned proximal segment of a tibia during a TPLO procedure and a distal portion extending along a longitudinal axis and configured to be positioned over a distal segment of the tibia during the TPLO procedure.

(4) The TPLO plate also includes a first distal hole extending through a proximal end of the distal portion of the body from the first surface to the second surface. The first distal hole includes an elongated portion and an offset portion open to and in communication with one another. The elongated portion extends along the longitudinal axis and an offset portion extending distally from the elongated portion along an offset axis, which extends at an angle relative to the longitudinal axis.

(5) In addition, the TPLO plate includes a second distal hole extending through the distal portion of the body distally of the first distal hole from the first surface to the second surface. The second distal hole extends along the longitudinal axis in alignment with the elongated portion of the first distal hole and including a sloped compression surface along a distal portion thereof. The first distal hole is configured such that when a first distal compression to the osteotomy cut is applied via insertion of a bone fixation element into a distal portion of the second distal hole, the bone fixation element translates along the offset axis from a central axis to a point of intersection to provide a cranial compression to the osteotomy cut.

(6) In an embodiment, a distance between a central axis of the offset portion of the first distal hole along which the bone fixation element is configured to be inserted and a point of intersection of the longitudinal axis and the offset axis is selected so that during the first distal compression, the bone fixation element received along the central axis is configured to move proximally along the offset axis from the central axis to the point of intersection.

(7) In an embodiment, an angle between the longitudinal axis along which the elongated portion extends and the offset axis A along which the offset portion extends corresponds to a desired cranial displacement.

(8) In an embodiment, the TPLO plate further includes a third distal hole extending through the distal portion of the body, between the first and second holes, from the first surface to the second surface and extending along the longitudinal axis, a distal portion of the third distal hole including a sloped compression surface.

(9) In an embodiment, the third distal hole is configured as a combi-hole including a proximal portion configured as a locking hole and a distal portion configured as a dynamic compression hole, the proximal and distal portions of the combi-hole open to and in communication with one another.

(10) In an embodiment, when a second distal compression is provided via the third distal hole, the bone fixation element received within the first distal hole translated proximally along the elongated portion of the first distal hole.

(11) In an embodiment, the second distal hole is configured as a dynamic compression hole including the sloped compression surface.

(12) In an embodiment, the proximal portion of the body includes three proximal holes, each of which extends through the proximal portion, from the first surface to the second surface, along a proximal edge thereof, each of the proximal holes being configured to receive a proximal bone fixation element therein.

(13) In an embodiment, a first one of the three proximal holes extends through the proximal portion in a position configured to facilitate insertion of a first one of the proximal bone fixation elements therethrough into a caudal portion of the proximal segment of the tibia. A second one of the proximal holes extends through the proximal portion in a position configured to facilitate insertion of a second one of the proximal bone fixation elements therethrough into a proximal portion of the proximal segment of the tibia. A third one of the proximal holes extends through the proximal portion in a position configured to facilitate insertion of a third one of the proximal bone fixation

elements therethrough into a cranial portion of the proximal segment of the tibia.

(14) In addition, the present disclosure relates to a Tibial Plateau Leveling Osteotomy (TPLO) plate for providing a cranial and a distal compression of an osteotomy cut. The TPLO plate includes a body extending longitudinally from a proximal end to a distal end and defined via a first surface which, in an operative configuration, faces away from a bone and a second surface which, in the operative configuration, faces toward the bone. The body includes a proximal portion configured to be positioned over a cut and repositioned proximal segment of a tibia during a TPLO procedure and a distal portion extending along a longitudinal axis and configured to be positioned over a distal segment of the tibia during the TPLO procedure.

(15) In addition, the TPLO plate includes a first distal hole extending through the distal portion of the body, proximate the distal end thereof, from the first surface to the second surface, the first distal hole including an elongated portion and an offset portion open to and in communication with one another. The elongated portion extends along the longitudinal axis and an offset portion extending distally from the elongated portion along an offset axis, which extends at an angle relative to the longitudinal axis.

(16) Furthermore, the TPLO plate includes a second distal hole extending through the distal portion of the body proximally of the first distal hole from the first surface to the second surface, the second distal hole extending along the longitudinal axis in alignment with the elongated portion of the first distal hole and including a sloped compression surface along a distal portion thereof. The first distal hole is configured such that when a first distal compression to the osteotomy cut is applied via insertion of a bone fixation element into a distal portion of the second distal hole, the bone fixation element translates along the offset axis from a central axis to a point of intersection to provide a cranial compression and a first distal compression to the osteotomy cut.

(17) In an embodiment, the plate further includes a third distal hole extending through the distal portion of the body, between the first hole and the second hole, from the first surface to the second surface and extending along the longitudinal axis, a distal portion of the third distal hole including a sloped compression surface.

(18) Furthermore, the present disclosure relates to a method for a Tibial Plateau Leveling Osteotomy (TPLO). The method includes positioning a bone plate in a desired initial position with a first surface of the bone plate facing away from a tibia and a second surface thereof facing a tibia so that a proximal portion of the bone plate extends over a proximal tibial segment and a distal portion of the bone plate extends over a distal tibial segment that has been cut away from the proximal tibial segment and rotated and seated within a recess formed in the distal tibial segment when the proximal tibia segment was cut away; inserting a first distal bone fixation element into the distal tibial segment of the tibia along a central axis of an offset portion of a first distal hole, which includes an elongated portion extending along a longitudinal axis of the distal portion of the bone plate and the offset portion extending distally from the elongated portion along an offset axis extending at an angle relative to the longitudinal axis; and inserting a second distal bone fixation element into the distal tibial segment via a second distal hole extending through the distal portion of the bone plate distally of the first distal hole so that a head portion of the second distal bone fixation element slides along a sloped compression surface extending along a distal portion of the second distal hole to move the bone plate distally relative to the second distal bone fixation element and provide a first distal compression between the cut and rotated proximal tibial segment, the first distal bone fixation element translating proximally along the offset axis during the first distal compression so that the bone plate rotates about the second distal bone fixation element and the proximal portion of the bone plate is moved in a cranial direction to provide a cranial compression of the proximal tibial segment against the distal tibial segment.

(19) In an embodiment, the second distal hole extends long the longitudinal axis, in alignment with the elongated portion of the first distal hole.

(20) In an embodiment, during the cranial compression, the first distal bone fixation element

translates along the offset axis from the central axis to a point at which the longitudinal axis and the offset axis intersect.
(21) In an embodiment, the method further includes inserting a third distal bone fixation element into the distal tibial segment via a third distal hole extending through the distal portion of the bone plate between the first and second distal holes so a head portion of the second distal bone fixation element slides along a sloped compression surface extending along a distal portion of the third distal hole of the third distal hole to move the bone plate distally relative to the third distal bone fixation element and provide a second distal compression between the cut and rotated proximal tibial segment and the distal tibial segment.
(22) In an embodiment, the first distal bone fixation element translates proximally along the elongated portion of the first distal hole.
(23) In an embodiment, the third distal bone fixation element extends along the longitudinal axis, in alignment with the second distal hole and the elongated portion of the first distal hole.
(24) In an embodiment, the method further includes inserting a first proximal bone fixation element through a first proximal hole and a second proximal bone fixation element through a second proximal hole into the proximal tibia segment, the first and second proximal holes extending through the proximal portion of the bone plate so that, when the bone plate is in the desired initial position, the first and second proximal bone fixation elements fix the proximal portion of the bone plate relative to the proximal tibial segment prior to insertion of the second distal bone fixation element through the second distal hole.
(25) In an embodiment, the first proximal bone fixation element is inserted into a cranial side of the proximal tibial segment.

Description

BRIEF DESCRIPTION
(1) FIG. **1** shows a plan view of a TPLO bone plate according to an exemplary embodiment of the present disclosure;
(2) FIG. **2** shows an enlarged plan view of a first hole extending through a distal portion of the TPLO plate of FIG. **1**;
(3) FIG. **3** shows a plan view of the TPLO plate of FIG. **1**, with a first a bone fixation element and a second bone fixation element inserted into an offset portion of the first hole and a second hole, respectively;
(4) FIG. **4** shows a plan view of the TPLO plate of FIG. **1**, after a first distal compression and a cranial compression;
(5) FIG. **5** shows a plan view of the TPLO plate of FIG. **1**, after a second distal compression; and
(6) FIG. **6** shows a plan view of a TPLO bone plate according to another exemplary embodiment of the present disclosure.
DETAILED DESCRIPTION
(7) The present disclosure may be further understood with reference to the following description and the appended drawings, wherein like elements are referred to with the same reference numerals. The present disclosure relates to a Tibial Plateau Leveling Osteotomy (TPLO) plate and, in particular, relates to a TPLO plate configured to provide both cranial (i.e., transverse) and distal (i.e., axial) compression during a TPLO procedure. Exemplary embodiments of the present disclosure describe a TPLO plate comprising a proximal portion configured to be positioned over a cut and repositioned upper portion (e.g., proximal portion) of a tibia, and a distal portion configured to be positioned along a lower portion (e.g., distal portion) of the tibia.
(8) The distal portion of the plate includes three holes positioned along the length of the distal portion so that bone fixation members inserted into these holes work in concert to provide both

cranial and distal compression of the interface between the cut and repositioned portion of the tibia and the lower portion of the tibia. The first hole includes an elongated portion extending along a longitudinal axis of the distal portion of the plate and an offset portion that extends distally therefrom, along an offset axis angled with respect to the longitudinal axis.

(9) A surface (i.e., a side wall) of the offset portion of the first hole meets a surface (i.e., a side wall) of the elongated portion of the first hole at an angle. In an embodiment, the second hole is elongated along the longitudinal axis and is distal of the first hole so that, when a first bone fixation element is inserted through the offset portion of the first hole into the bone and a second bone fixation element is inserted into the bone via the second hole, sliding the plate distally relative to the first and second bone fixation elements, moves the plate distally while the proximal portion of the plate is rotated due to the angulation of the offset axis until the first bone fixation element enters the elongated portion of the first hole. As the plate is moved so that distally from this point, no further rotation of the plate occurs. The angulation of the offset axis with respect to the longitudinal axis is selected so that the rotation of the plate as the first bone fixation element traverses the offset portion moves the proximal portion of the plate (and the cut and rotated section of bone coupled thereto) relative to the lower portion of the tibia to provide cranial compression at the interface between the cut portion of bone and the lower tibia while movement of the plate distally relative to the first and second bone fixation elements moves the proximal portion of the plate distally to apply a first distal compression across the osteotomy cut.

(10) The distal portion of the plate further includes a third hole extending therethrough, between the first and second holes. The third hole is configured so that, when engaged via a third bone fixation element inserted therethrough, the plate is moved distally relative to the tibia so that the position of the first bone fixation element within the elongated portion of the first hole is increasingly proximal, and a second distal compression is provided across the osteotomy cut.

(11) It will be understood by those of skill in the art that although the TPLO plates of the present embodiments are described with respect to a canine CCL, the TPLO plate of the present disclosure may also be used to treat the tibias of other quadrupeds such as, for example, felines, bovines, equines, etc. It will also be understood by those of skill in the art that the terms proximal, distal, caudal, and cranial, as used herein, are anatomical directional terms for an animal such as, for example, a canine and are employed in a manner consistent with their standard anatomical meanings. In particular, the term proximal refers to a direction toward a center of the body while distal refers to a direction away from the center of the body while cranial refers to a direction toward a head of the animal and caudal refers to a direction away from the head.

(12) As shown in FIGS. **1-5**, a TPLO plate **100** according to an exemplary embodiment of the present disclosure is configured to secure proximal and distal segments of a tibia which have been separated from one another during a TPLO procedure via a substantially curvilinear osteotomy cut as would be understood by those skilled in the art. During a TPLO procedure, a proximal segment of the tibia is cut away from the rest of the tibia, rotated relative to the tibia to a new position selected to enhance the stability of, for example, a canine stifle joint (e.g., after injury to a cranial cruciate ligament (CCL)), and secured to the distal segment in this desired position—e.g., via a bone plate coupled to the cut segment of the tibia and the distal segment of the tibia.

(13) According to an exemplary embodiment, the plate **100** comprises a body **101** including a proximal portion **102** sized and shaped to be positioned over and coupled to the cut and repositioned proximal segment of the tibia, and a distal portion **104** sized and shaped to extend along and be coupled to the distal segment of the tibia so that the plate **100** fixes the position of the cut and repositioned portion of the tibia relative to the distal segment. In an exemplary embodiment the distal portion **104** includes at least three holes—a first hole **106**, a second hole **108**, and a third hole **110**—extending therethrough.

(14) Those skilled in the art will understand that additional holes may be added as desired so long as they do not impede the functioning of the plate **100** with respect to the distal and cranial

compression as will be described below. The first hole **106** includes an elongated portion **112** extending along a longitudinal axis L of the plate **100** and an offset portion **114** extending distally from the elongated portion **112** along an offset axis A that is angled with respect to the longitudinal axis L. As will be described in further detail below, the offset portion **114** of the first hole **106** and the second hole **108** are configured to work together so that, when first and second bone fixation elements **116**, **118** are inserted therein, respectively, desired levels of distal and cranial compression of the osteotomy cut are provided at the interface between the cut and repositioned portion of the bone and the distal segment of the tibia.

(15) The elongated portion **112** of the first hole **106**, the second hole and the third hole **110** are configured to work together so that, when a third bone fixation element **120** is subsequently inserted into the third hole **110**, a desired amount of additional distal compression is provided. The amounts of cranial and distal compression are selected to ensure that the cut and repositioned proximal segment of the tibia is seated within a recessed portion of the distal segment formed via the curvilinear cut to optimize healing of the bone.

(16) As shown in FIG. **1**, the plate **100** includes a body **101** extending longitudinally from a proximal end **122** to a distal end **124**. The body **101** is defined via a first surface **126** which, in an operative configuration, faces away from a bone (e.g., the tibia), and a second surface **128** which, in the operative configuration, faces toward the bone. The body **101** includes the proximal portion **102** and the distal portion **104**, which are connected to one another via a neck portion **130** so that, when in the operative configuration, the proximal portion **102** is positioned over and coupled to the proximal segment of the tibia, as desired, and the distal portion **104** extends over and is coupled to the distal segment of the tibia, the neck portion **130** extends across the interface between the cut-away portion of the proximal tibia and the distal tibia at the curvilinear osteotomy cut.

(17) The distal portion **104** extends distally of the proximal portion **102** along the longitudinal axis L. According to an exemplary embodiment, the neck portion **130** of one embodiment is curved so that the proximal portion **102** is offset (e.g., angled) with respect to the longitudinal axis L along which the distal portion **104** extends. For example, the proximal portion **102** may be angled in a caudal direction relative to the longitudinal axis L although, as would be understood by those of skill in the art, the angle may be selected in any manner desired to conform to the geometry of the cut-away and rotated portion of the tibia relative to the distal tibia (or any other bone segment involved) in any given procedure.

(18) As will be understood by those of skill in the art, the proximal portion **102** is, in this embodiment, preferably constructed, sized, shaped, and contoured to conform to the shape and orientation of the cut-away proximal segment of the tibia when the cut-away segment is in a desired position relative to the distal tibia and the distal portion **104** is in a desired position on the distal tibia. In particular, the second surface **128** of this embodiment is specifically contoured so that, when the proximal portion **102** is positioned over the proximal segment of the tibia, the second surface **128** extends along an exterior surface of the cut and rotated proximal segment of the tibia, in contact therewith.

(19) According to an exemplary embodiment, the proximal portion **102** includes at least three holes —a first hole **132**, a second hole **134**, and a third hole **136**—each of which extends through the proximal portion **102** along a central axis, from the first surface **126** to the second surface **128**. The first, second and third holes **132**, **134**, **136** may be positioned adjacent an edge extending along the proximal end **122** of the plate **100**. Each of the first, second and third holes **132**, **134**, **136** of the proximal portion **102** of this embodiment is configured to receive therein a bone fixation element such as, for example, a locking screw to fix the proximal portion **102** of the plate **100** relative to the cut and rotated proximal segment of the tibia.

(20) In an exemplary embodiment, the first hole **132** is positioned on the plate **100** and oriented so that, when the plate **100** is positioned on a tibia in a desired position, the first hole **132** is positioned to receive a bone fixation element through a caudal portion of the resected proximal segment of the

tibia, while the second hole **134** is positioned and oriented to receive a bone fixation element through a proximal portion of the resected proximal segment of the tibia, and the third hole **136** is positioned and oriented on the plate **100** to receive a bone fixation element through a cranial portion of the resected proximal segment of the tibia. As would be understood by those skilled in the art, central axes of each of the first, second and third holes **132**, **134**, **136** may be optionally angled to direct bone fixation elements inserted therealong into a desired portion of the proximal segment of the bone (e.g., a central mass of the resected portion of the proximal tibia).
(21) Furthermore, any or all of the first, second and third holes **132**, **134**, **136** of the proximal portion **102** may be configured as locking holes, including a threading extending therein for engaging corresponding threading on the head of a bone fixation element inserted therein. Thus, bone fixation elements inserted therein may be locking screws including corresponding threading along a head portion thereof. It will be understood by those of skill in the art, however, that the first, second and third holes **132**, **134**, **136** of the proximal portion **102** of the plate **100** may have any of a variety of configurations so long as bone fixation elements are insertable therethrough to be inserted into a desired portion of the bone.
(22) The distal portion **104** of the plate **100** is contoured to extend, when in the desired position, along the distal segment of the tibia. In particular, in the operative configuration, the second surface **128** extends along the distal segment, in contact therewith. As described above, the distal portion **104** of this embodiment also at least includes three holes extending therethrough—the first hole **106**, the second hole **108** and the third hole **110**. The first hole **106**, the second hole **108**, and the third hole **110** work in concert to provide desired levels of cranial and distal compression to the osteotomy cut of the tibia.
(23) The first hole **106** is positioned adjacent to a proximal end **138** of the distal portion **104**, adjacent to the neck portion **130**. The first hole **106** extends through the plate **100** from the first surface **126** to the second surface **128** and includes the elongated portion **112** and the offset portion **114**, each of which is configured to receive the first bone fixation element **116** therein. The elongated portion **112** is elongated along the longitudinal axis L while the offset portion **114** extends distally therefrom along the offset axis A, which is angled with respect to the longitudinal axis L. The elongated portion **112** and the offset portion **114** are open to and in communication with one another so that a surface **140** (i.e., a side wall of the elongated portion **112**) defining the elongated portion **112** of the first hole **106** meets a surface **142** (i.e., a side wall of the offset portion **114**) defining the offset portion **114** at an angle corresponding to an angle B between the longitudinal axis L and the axis A.
(24) In an exemplary embodiment, the offset portion **114** extends distally from the elongated portion **112** angled cranially relative to the longitudinal axis L (i.e., so that the distal end of the offset portion **114** is separated transversely from the axis L on the cranial side of the axis L while the offset axis A, if extended proximally beyond the axis L would extend toward the caudal side. Thus, when the first bone fixation element **116** is received in the offset portion **114** and a second bone fixation element **118** is inserted through the second hole **108** and the plate **100** is moved distally over the first and second bone fixation elements **116**, **118**, respectively, the first and second bone fixation elements **116**, **118** remain stationary (i.e., coupled to the bone) while the plate **100** translates distally so that the first bone fixation element **116** passes proximally along the offset portion **114** while the portion of the plate **100** surrounding the second hole **108** moves distally along the axis L. This rotates the plate **100** so that the proximal end **122** of the plate moves in a cranial direction against the surface of the osteotomy cut in the distal segment of the tibia to apply cranial compression across the osteotomy cut, as will be described in further detail below.
(25) The surfaces **140**, **142** defining the first hole **106** are, in this embodiment, configured to correspond to a size of a head portion of the first bone fixation element **116**. For example, the surfaces **140**, **142** may taper from the first surface **126** toward the second surface **128** and/or may include a curvature corresponding to an underside the head portion (e.g., a surface of the head

portion which is configured to engage the first and second surfaces **126**, **128**) so that upon insertion of the first bone fixation element **116** thereinto, the head portion of the first bone fixation element **116** is slidable along the axes of the elongated portion **112** and the offset portion **114**, as will be described in further detail below.

(26) In addition, the first hole **106** of this embodiment is configured with the length of the elongated portion **112** selected so that, when the first bone fixation element was initially positioned as desired, the head portion of the first bone fixation element **116** is, after the final distal compression, seated along the surface **140** to sit flush with the first surface **126** of the plate **100**, as will be understood by those of skill in the art. In an exemplary embodiment, the first bone fixation element **116** is, for example, a standard cortex screw.

(27) In an exemplary embodiment, the angle B between the longitudinal axis L and the offset axis A may range from between approximately 10 degrees to 35 degrees. The offset portion **114** of the first hole **106**, however, may have any of a variety of configurations and angulations relative to the elongated portion **112** depending on the positions of the first and second holes along the axis L and a desired amount of cranial and distal compression.

(28) As would be understood by those skilled in the art, the angle B is selected based on a desired cranial compression to be provided by the plate **100** for a given amount of distal compression. A distance D between a central axis C of the offset portion **114** along which the first bone fixation element **116** is configured to be inserted and a point of intersection P of the longitudinal axis L and the offset axis A is specifically selected such that a first distal compression achieved via insertion of the second bone fixation element **118** into the second hole **108** will translate the first bone fixation element **116** received along the central axis C proximally along the axis A from this initial position along the central axis C to the point of intersection P.

(29) It will be understood by those or skill in the art that the distance D and the angle B between the axis L and offset axis A may be determined based on any of a number of factors including, for example, a desired cranial and distal compression, and a distance between various holes **106**, **108**, **110**. That is, those skilled in the art will understand the geometric relationships that allow a plate **100** to be designed to provide a desired amount of rotation of the proximal portion of the plate **100** (cranial compression) for a given amount of distal compression.

(30) The second hole **108** extends through the distal portion **104**, distally of the first hole **106**, and is configured to receive the second bone fixation element **118** therein. In an exemplary embodiment, the second hole **108** extends through the distal portion **104** proximate the distal end **124** and in longitudinal alignment with the elongated portion **112** of the first hole **106**. The second hole **108** in this embodiment is configured as a dynamic compression hole configured to provide distal compression across the osteotomy cut. For example, the second hole **108** of this embodiment extends through the distal portion **104** from the first surface **126** to the second surface **128** and is elongated along the longitudinal axis L, in alignment with the elongated portion **112** of the first hole **106**. The second hole **108** includes a proximal portion **144** and a distal portion **146**, which facilitates translational movement of second hole **108** relative to the second bone fixation element **118** that is received therein so that the second bone fixation element traverses from the distal portion **146** toward the proximal portion **144**, as a first distal compression is applied to the tibia.

(31) The distal portion **146** of the second hole **108** includes a sloped compression surface **148** inclined at a curve/angle selected so that, as an underside of a head portion of the second bone fixation element **118** is pressed thereagainst (e.g., as the third bone fixation element **120** is inserted gradually further into the bone and further through the plate **100**), the sloped compression surface **148** slides along the head portion of the second bone fixation element **118**. In particular, as the second bone fixation element **118** is driven more deeply into the bone, the head portion of the second bone fixation element **118** slides along the sloped compression surface **148** moving the plate **100** distally relative to the second bone fixation element **118**, thereby applying distal compression across the osteotomy—i.e., pulling the proximal segment of the tibia distally against

the distal segment of the tibia as the proximal portion **102** of the plate **100**, which is coupled to the cut-away proximal segment of the tibia, is drawn distally by the movement of the distal portion of the plate **100**.

(32) According to an exemplary embodiment, the proximal portion **144** of the second hole **108** is defined via a surface **150** sized, shaped, and configured to correspond to the size and shape of an underside of the second bone fixation element **118** which, in one embodiment, is a cortex screw. In an exemplary embodiment, the surface **150** may be tapered and/or curved from the first surface **126** toward the second surface **128** and, in one example, is configured as a spherical relief configured to seat the head portion of the second bone fixation element **118** therein so that the head portion of the second bone fixation element **118** is substantially flush with the first surface **126** of the body **101** of the plate **100** as would be understood by those skilled in the art. It will also be understood by those of skill in the art, however, that the head portion of the second bone fixation element **118** is not required to be finally seated within the spherical relief as the plate **100** moves so that the second bone fixation element **118** passes from the distal portion **146** toward the proximal portion **144** via a distance corresponding to a desired axial compression.

(33) As discussed above, the offset portion **114** of the first hole **106** and the second hole **108** work in concert so that, when the first bone fixation element **116** and the second bone fixation element **118** are inserted into respective ones of the first and second holes **106**, **108**, cranial compression is applied to the osteotomy cut. In particular, the first bone fixation element **116** may be inserted into the offset portion **114** along the central axis C thereof and the second bone fixation element **118** is subsequently inserted into the distal portion **146** of the second hole **108** so that the head portion of the second bone fixation element **118** with the sloped compression surface **148** moves the plate **100** distally relative to the second bone fixation element **118**. This first distal compression is applied until the plate **100** has moved so that the first bone fixation element **116** has translated along the offset axis A from the central axis C to the point of intersection P, where the longitudinal axis L of the elongated portion **112** of the first hole **106** and the offset axis A of the offset portion **114** intersect.

(34) Further insertion of the second bone fixation element **118** into the bone translates the first bone fixation element **116** proximally along the offset axis A as the first bone fixation element **116** interfaces with the surface **142** of the offset portion **114**. This causes the plate **100** to rotate about the second bone fixation element **118** (clockwise as seen in FIG. **4**) so that the proximal portion **102** of the plate **100**, which is coupled to the cut proximal segment of the tibia, moves in a cranial direction, applying a cranial compression across the osteotomy cut. Thus, this portion of the first distal compression simultaneously applies the cranial compression.

(35) The third hole **110**, extends through the distal portion **104** in a position extending between the first and second holes **106**, **108**, and is configured to receive the third bone fixation element **120** therein. In an exemplary embodiment, the third hole **110** extends through the body **101** from the first surface **126** to the second surface **128** along the longitudinal axis L, substantially in alignment with the elongated portion **112** of the first hole **106** and the second hole **108**. In one exemplary embodiment, the third hole **110** is configured as a combi-hole including a proximal portion **152** configured to provide locking fixation of the plate **100** relative to the bone and a distal portion **154** configured to provide dynamic compression. The proximal and distal portions **152**, **154** are open to and in communication with one another.

(36) The proximal portion **152** in this embodiment is configured as a locking hole extending through the first surface **126** to the second surface **128** along a central axis, which extends perpendicular relative to the longitudinal axis L. In one embodiment, the central axis of the proximal portion **152** extends substantially perpendicular to the longitudinal axis L. It will be understood by those of skill in the art, however, that the central axis of the proximal portion **152** may be non-perpendicularly angled relative to the central axis of the proximal portion **152** so that a bone fixation element inserted therealong may be inserted into a desired portion of a bone offset

from the portion immediately beneath the axis L. A surface **156** defining the proximal portion **152** may include a threading extending therealong for engaging a corresponding threading along a head of, for example a locking screw, to provide a locking fixation between the plate and the bone.

(37) Similarly to the second hole **108**, the distal portion **154** in this embodiment includes a sloped compression surface **158** inclined at a curve/angle selected so that, as an underside of a head portion of the third bone fixation element **120** is pressed thereagainst (e.g., as the third bone fixation element **120** is inserted gradually further into the bone and further through the distal portion **154**), the sloped compression surface **158** slides along the head portion of the third bone fixation element **120**. As the third bone fixation element **120** is driven further into the bone, the head portion of the third bone fixation element **120** slides along the sloped compression surface **158** moving the plate **100** distally relative to the third bone fixation element **120**. In this embodiment, the third bone fixation element **120** is a standard cortex screw.

(38) As discussed above, the elongated portion **112** of the first hole **106** and the third hole **110** work in concert so that, when the third bone fixation element **120** is subsequently inserted into the distal portion **154** of the third hole **110**, the head portion of the third bone fixation element **120** slides along the sloped compression surface **158** providing a second distal compression across the osteotomy cut, drawing the cut proximal segment of the tibia which is coupled to the proximal portion **102** of the plate, further distally against the distal segment of the tibia. As the plate **100** is moved distally relative to the third bone fixation element **120**, the first bone fixation element **116** translates proximally along the elongated portion **112** of the first hole **106**.

(39) According to an exemplary method, the plate **100** may be used to provide desired amounts of both cranial and distal compression during a TPLO procedure. As will be understood by those of skill in the art, upon cutting of a proximal segment of a tibia from a distal segment of the tibia via a substantially curvilinear osteotomy cut, the proximal segment is rotated and repositioned relative to the distal segment and the plate **100** is then placed over the separated bone segments so that the proximal portion **102** is positioned over the cut and repositioned proximal segment of the tibia in a desired alignment and the distal portion **104** is positioned over a target portion of the distal segment of the tibia.

(40) According to an exemplary method, the first bone fixation element **116** is inserted along the central axis C of the offset portion **114** of the first hole **106** and the second bone fixation element **118** is inserted through the distal portion **146** of the second hole **108** to establish a preliminary position of the plate **100** relative to the tibia so that the proximal portion **102** of the body **101** of the plate **100** is positioned over the cut proximal segment of the tibia and the distal portion **104** is positioned over the distal segment of the tibia. Once the preliminary position has been established, bone fixation elements may be inserted through at least two of the holes extending through the proximal portion **102** of the plate **100** to fix the proximal portion **102** relative to the proximal segment of the tibia. In one embodiment, bone fixation elements are inserted through a cranial one of the holes (e.g., third hole **136**) and one of the other holes (e.g., the first hole **132** and the second hole **134**) of the proximal portion **102**.

(41) The second bone fixation element **118** may then be tightened (e.g., rotatably inserted further into the distal portion **146** of the second hole **108** so that the head portion of the second bone fixation element **118** engages the compression surface **148** thereof, causing the plate **100** to be moved distally with respect to the second bone fixation element **118** to apply a first distal compression to the osteotomy cut—i.e., by moving the proximal segment of the tibia distally toward the distal segment of the tibia. As the plate **100** is being moved distally with respect the second bone fixation element **118** (i.e., the second bone fixation element **118** is translated from the distal portion **146** toward the proximal portion **144** of the second hole **108**), the first bone fixation element **116** also translates along the offset axis A of the offset portion **114** of the first hole **106**, rotating the plate **100** about the second bone fixation element **118** and providing a simultaneous cranial compression in which the proximal portion **102** of the plate **100** is moved in a cranial

direction.
(42) In particular, as the plate **100** moves so that the first bone fixation element **116** translates proximally along the offset axis A, the first bone fixation element **116** interfaces with a portion of the surface **142**, which forces the proximal portion **102** of the plate **100** in the cranial direction.
(43) As described above, the movement of the plate **100** generating the first distal compression moves the plate relative to the first bone fixation element **116** so that the first bone fixation element **116** translates from its insertion position along the central axis C to the point P at which the longitudinal axis L of the elongated portion **112** intersects the offset axis A. The third bone fixation element **120** may then be inserted into the distal portion **154** so that the head portion of the third bone fixation element **120** engages the compression surface **158** thereof, moving the plate **100** further distally with respect to the distal segment of the tibia to apply a second distal compression across the osteotomy cut. During the second distal compression the plate **100** moves so that the third bone fixation element **120** translates from the distal portion **154** toward the proximal portion **152** of the third hole **110** and the first bone fixation element **116** translates from the point P along the longitudinal axis L of the elongated portion **112** of the first hole **106** toward a proximal end **160** thereof.
(44) Upon completion of the cranial and distal compression, the first bone fixation element **116** is tightened to fix the distal portion **104** of the plate **100** relative to the distal segment of the tibia. Additional bone fixation elements may be inserted through any remaining holes extending through to provide final fixation as desired. For example, bone fixation elements may be inserted through any remaining holes **132**, **134**, **136** of the proximal portion **102**, to provide further locking of the plate **100** in the desired position relative to the tibia.
(45) Except as detailed below, a TPLO plate **200** according to another exemplary embodiment shown in FIG. **6**, is substantially similar to the plate **100**, as described above, comprising a proximal portion (not shown) and a distal portion **204** configured to fix a proximal segment of a tibia relative to a distal segment of the tibia, respectively, during a TPLO procedure. The proximal portion is substantially similar to the proximal portion **102** of the plate **100** and is sized and shaped to be positioned over and coupled to the proximal segment of the tibia, which is cut away from the distal segment and repositioned relative thereto. The distal portion **204** is sized and shaped to extend along and be coupled to the distal segment (i.e., the shaft) of the tibia so that the plate **200** fixes the position of the cut and repositioned portion of the tibia relative to the distal segment. Similarly to the plate **100**, the distal portion **204**, includes at least three holes—a first hole **206**, a second hole **208**, and a third hole **210**, configured to work together to provide distal and cranial compression across the osteotomy cut and ensure that the cut and repositioned proximal segment of the tibia is optimally seated within a recessed portion of the distal segment formed via the osteotomy cut.
(46) Similarly to the first hole **106**, the first hole **206** includes an elongated portion **212** extending along a longitudinal axis 2L of the plate **200** (along which the distal portion **204** extends) and an offset portion **214** extending distally from the elongated portion **212** along an offset axis A angled with respect to the longitudinal axis L. The offset portion **214** of the first hole **206** and the second hole **208** are configured to work together so that, when first and second bone fixation elements are inserted therein, respectively, desired levels of distal and cranial compression of the osteotomy cut are provided at the interface between the cut and repositioned portion of the bone and the distal segment of the tibia. The elongated portion **212** of the first hole **206**, the second hole and the third hole **210** are configured to work together so that, when a third bone fixation element is subsequently inserted into the third hole **210**, a desired amount of additional distal compression is provided.
(47) As will be described in further detail below, however, a position of the first and second holes **206**, **208** along the distal portion **204** is transposed relative to the plate **100**, and an offset portion **214** of the first hole **206** extends distally from an elongated portion **212** of the first hole **206** at an

angle extending in a caudal direction—i.e., so that the first hole **206** is a mirror image of the first hole **106** relative to the longitudinal axis **2**L.
(48) The first hole **206**, in this embodiment, is positioned adjacent to a distal end **224** of the distal portion **204**. Similarly to the first hole **106**, the first hole **206** extends through the distal portion **204** and includes the elongated portion **212** and the offset portion **214**, which are open to one another. The elongated portion **212** extends along the longitudinal axis **2**L while the offset portion **214** extends distally from the elongated portion **212**, along an offset axis **2**A angled with respect to the longitudinal axis **2**L. As discussed above, the first hole **206** is substantially similar to the first hole **106**, as described above with respect to the plate **100**, except that the first hole **206** is a mirror image of the first hole **106**, as described above with respect to the plate **100**, relative to the longitudinal axis **2**L. In particular, the offset portion **214** extends distally from the elongated portion **212**, angled caudally relative to the longitudinal axis **2**L (i.e., so that the distal end of the offset portion is separated transversely from the axis **2**L on a caudal side of the axis **2**L while the offset axis **2**A, if extends proximally beyond the axis **2**L would extend toward a cranial side).
(49) As will be described in further detail below, when a first bone fixation element is received within the offset portion **214** and the plate **200** is moved in distally relative to the first bone fixation element, the first bone fixation element moves through the offset portion **214**, rotating of the plate **200** so that the distal end **224** of the plate **200** is moved in a caudal direction and a proximal end of the plate is moved in a cranial direction to apply a cranial compression across the osteotomy cut.
(50) The second hole **208** is substantially similar to the second hole **108** described with respect to the plate **100**. The second hole **208**, however, extends through the distal portion **204** proximally of the first hole **206** and, in one exemplary embodiment, is positioned proximate a proximal end **238** of the distal portion **204**. Similarly to the second hole **108**, the second hole **208** extends through the distal portion **204** in longitudinal alignment with the elongated portion **212** of the first hole **206** and, in this embodiment, is configured as a dynamic compression hole.
(51) In particular, the second hole **208** includes a proximal portion **244** and a distal portion **246** which includes a sloped compression surface facilitating translational movement of the second hole **208** relative to a second bone fixation element received therein, so that the second bone fixation element traverses from the distal portion **246** toward the proximal portion **244**, a first distal compression is applied to the tibia. As the second bone fixation element is driven into the bone, a head portion thereof slides along the sloped compression surface of the distal portion **246** of the second hole **208**, moving the plate **200** distally relative to the second bone fixation element so that a distal compression is applied across the osteotomy.
(52) Similarly to the plate **100**, the offset portion **214** of the first hole **206** and the second hole **208** work in concert so that, when the first bone fixation element and the second bone fixation element are inserted into respective ones of the first and second holes **206**, **208**, cranial compression is applied to the osteotomy cut. In particular, the first bone fixation element may be inserted into the offset portion **214** and the second bone fixation element subsequently inserted into the distal portion **246** of the second hole **208** so that the head portion of the second bone fixation element engages the sloped compression surface of the distal portion **246** to move the plate **200** distally relative to the second bone fixation element.
(53) This first distal compression is applied until the plate **200** has moved so that the first bone fixation element has translated along the offset axis **2**A to a point where the longitudinal axis **2**L and the offset axis **2**A intersect. Since the offset axis **2**A is angled with respect to the longitudinal axis **2**L, the distal translation of the plate **200** relative to first bone fixation element rotates the plate **200** about the second bone fixation element such that the distal end **224** of the plate **200** moves in a caudal direction while the proximal end of the plate **200** moves in a cranial direction. Thus, the proximal portion of the plate **200**, which is coupled to the proximal segment of the tibia, moves in the cranial direction, applying a cranial compression across the osteotomy cut. Accordingly, the first distal compression achieved via the insertion of the second bone fixation element into the

second hole **208** applies a simultaneous cranial compression.

(54) The third hole **210** is substantially similar to the third hole **110**, described above with respect to the plate **100**. In particular, the third hole **210** extends through distal portion **204** in a position extending between the first and second holes **206**, **208**, in longitudinal alignment therewith. In one embodiment, the third hole **208** is configured as a combi-hole. A distal portion **254** of the third hole **208** is configured to provide dynamic compression and may include a sloped compression surface. Thus, after achieving a cranial compression and a first level of distal compression via the insertion of the first and second bone fixation elements through the offset portion **214** of the first hole **206** and the distal portion **246** of the second hole **208**, a third bone fixation element may be inserted through the distal portion **254** to provide a second distal compression.

(55) As discussed above, the elongated portion **212** of the first hole **206** and the third hole **210** work in concert so that, when the third bone fixation element is subsequently inserted into the distal portion **254** of the third hole **210**, the head portion of the third bone fixation element slides along the sloped compression surface of the distal portion **254** to move the plate **200** distally relative to the third bone fixation element. As the plate **200** is moved distally relative to the third bone fixation element, the first bone fixation element received within the first hole **206** translates proximally (relative to the plate **200**) along the elongated portion **212** of the first hole **206**. Thus, the cut proximal segment of the tibia, which is coupled to the proximal portion of the plate **200**, is drawn further distally against the distal segment of the tibia.

(56) As will be understood by those of skill in the art, the plate **200** may be used in a manner substantially similar to the plate **100** to provide both a cranial and a distal compression to an osteotomy cut. As described above, the first bone fixation element will be inserted through the offset portion **214** of the first hole **206** at the distal end **224** of the distal portion **204** of the plate **200**. The second bone fixation element will subsequently be inserted through the distal portion **246** of the second hole **208** proximate the proximal end **238** of the distal portion **204** of the plate **200** to provide the first distal compression and the cranial compression. The third bone fixation element may then be inserted through the distal portion **254** of the third hole **210** to provide the second, final distal compression.

(57) It will be understood by those of skill in the art that modifications and variations may be made in the structure and methodology of the present invention, without departing from the spirit and scope of the invention. Thus, it is intended that the present invention cover the modifications and variations of this invention, provided that they come within the scope of the appended claims and their equivalents.

Claims

1. A Tibial Plateau Leveling Osteotomy (TPLO) plate for providing a cranial and a distal compression of an osteotomy cut, comprising: a body extending longitudinally from a proximal end to a distal end and defined via a first surface which, in an operative configuration, faces away from a bone and a second surface which, in the operative configuration, faces toward the bone, the body including a proximal portion configured to be positioned over a cut and repositioned proximal segment of a tibia during a TPLO procedure and a distal portion extending along a longitudinal axis and configured to be positioned over a distal segment of the tibia during the TPLO procedure; a first distal hole extending through a proximal end of the distal portion of the body from the first surface to the second surface, the first distal hole including an elongated portion and an offset portion open to and in communication with one another, the elongated portion extending along the longitudinal axis and the offset portion extending distally from the elongated portion along an offset axis, which extends at an angle relative to the longitudinal axis; and a second distal hole extending through the distal portion of the body distally of the first distal hole from the first surface to the second surface, the second distal hole extending along the longitudinal axis in alignment

with the elongated portion of the first distal hole and including a sloped compression surface along a distal portion thereof, wherein the first distal hole is configured such that when a first distal compression to the osteotomy cut is applied via insertion of a bone fixation element into the distal portion of the second distal hole, a second bone fixation element translates along the offset axis from a central axis to a point of intersection to provide a cranial compression to the osteotomy cut.

2. The plate of claim 1, wherein a distance between the central axis of the offset portion of the first distal hole along which the second bone fixation element is configured to be inserted and the point of intersection of the longitudinal axis and the offset axis is selected so that during the first distal compression, the second bone fixation element received along the central axis is configured to move proximally along the offset axis from the central axis to the point of intersection.

3. The plate of claim 1, wherein the angle between the longitudinal axis along which the elongated portion extends and the offset axis along which the offset portion extends corresponds to a desired cranial displacement.

4. The plate of claim 1, further comprising: a third distal hole extending through the distal portion of the body, between the first and second distal holes, from the first surface to the second surface and extending along the longitudinal axis, a distal portion of the third distal hole including a sloped compression surface.

5. The plate of claim 4, wherein the third distal hole is configured as a combi-hole including a proximal portion configured as a locking hole and a distal portion configured as a dynamic compression hole, the proximal and distal portions of the combi-hole open to and in communication with one another.

6. The plate of claim 4, wherein when a second distal compression is provided via the third distal hole, the bone fixation element received within the first distal hole is translated proximally along the elongated portion of the first distal hole.

7. The plate of claim 1, wherein the second distal hole is configured as a dynamic compression hole including the sloped compression surface.

8. The plate of claim 1, wherein the proximal portion of the body includes three proximal holes, each of which extends through the proximal portion, from the first surface to the second surface, along a proximal edge thereof, each of the proximal holes being configured to receive a proximal bone fixation element therein.

9. The plate of claim 8, wherein a first one of the three proximal holes extends through the proximal portion in a position configured to facilitate insertion of a first one of the proximal bone fixation elements therethrough into a caudal portion of the proximal segment of the tibia, wherein a second one of the proximal holes extends through the proximal portion in a position configured to facilitate insertion of a second one of the proximal bone fixation elements therethrough into a proximal portion of the proximal segment of the tibia, and wherein a third one of the proximal holes extends through the proximal portion in a position configured to facilitate insertion of a third one of the proximal bone fixation elements therethrough into a cranial portion of the proximal segment of the tibia.

10. A Tibial Plateau Leveling Osteotomy (TPLO) plate for providing a cranial and a distal compression of an osteotomy cut, comprising: a body extending longitudinally from a proximal end to a distal end and defined via a first surface which, in an operative configuration, faces away from a bone and a second surface which, in the operative configuration, faces toward the bone, the body including a proximal portion configured to be positioned over a cut and repositioned proximal segment of a tibia during a TPLO procedure and a distal portion extending along a longitudinal axis and configured to be positioned over a distal segment of the tibia during the TPLO procedure; a first distal hole extending through the distal portion of the body, proximate the distal end thereof, from the first surface to the second surface, the first distal hole including an elongated portion and an offset portion open to and in communication with one another, the elongated portion extending along the longitudinal axis and the offset portion extending distally from the elongated portion

along an offset axis, which extends at an angle relative to the longitudinal axis; and a second distal hole extending through the distal portion of the body proximally of the first distal hole from the first surface to the second surface, the second distal hole extending along the longitudinal axis in alignment with the elongated portion of the first distal hole and including a sloped compression surface along a distal portion thereof, wherein the first distal hole is configured such that when a first distal compression to the osteotomy cut is applied via insertion of a bone fixation element into the distal portion of the second distal hole, a second bone fixation element translates along the offset axis from a central axis to a point of intersection to provide a cranial compression and the first distal compression compression to the osteotomy cut.

11. The plate of claim 10, further comprising: a third distal hole extending through the distal portion of the body, between the first distal hole and the second distal hole, from the first surface to the second surface and extending along the longitudinal axis, a distal portion of the third distal hole including a sloped compression surface.

12. A method for a Tibial Plateau Leveling Osteotomy (TPLO), comprising: positioning a bone plate in a desired initial position with a first surface of the bone plate facing away from a tibia and a second surface thereof facing the tibia so that a proximal portion of the bone plate extends over a proximal tibial segment and a distal portion of the bone plate extends over a distal tibial segment that has been cut away from the proximal tibial segment and rotated and seated within a recess formed in the distal tibial segment when the proximal tibia segment was cut away; inserting a first distal bone fixation element into the distal tibial segment of the tibia along a central axis of an offset portion of a first distal hole, which includes an elongated portion extending along a longitudinal axis of the distal portion of the bone plate and the offset portion extending distally from the elongated portion along an offset axis extending at an angle relative to the longitudinal axis; and inserting a second distal bone fixation element into the distal tibial segment via a second distal hole extending through the distal portion of the bone plate distally of the first distal hole so that a head portion of the second distal bone fixation element slides along a sloped compression surface extending along a distal portion of the second distal hole to move the bone plate distally relative to the second distal bone fixation element and provide a first distal compression between the cut and rotated proximal tibial segment, the first distal bone fixation element translating proximally along the offset axis during the first distal compression so that the bone plate rotates about the second distal bone fixation element and the proximal portion of the bone plate is moved in a cranial direction to provide a cranial compression of the proximal tibial segment against the distal tibial segment.

13. The method of claim 12, wherein the second distal hole extends long the longitudinal axis, in alignment with the elongated portion of the first distal hole.

14. The method of claim 12, wherein, during the cranial compression, the first distal bone fixation element translates along the offset axis from the central axis to a point at which the longitudinal axis and the offset axis intersect.

15. The method of claim 12, further comprising: inserting a third distal bone fixation element into the distal tibial segment via a third distal hole extending through the distal portion of the bone plate between the first and second distal holes so the head portion of the second distal bone fixation element slides along a sloped compression surface extending along a distal portion of the third distal hole to move the bone plate distally relative to the third distal bone fixation element and provide a second distal compression between the cut and rotated proximal tibial segment and the distal tibial segment.

16. The method of claim 15, wherein the first distal bone fixation element translates proximally along the elongated portion of the first distal hole.

17. The method of claim 15, wherein the third distal bone fixation element extends along the longitudinal axis, in alignment with the second distal hole and the elongated portion of the first distal hole.

18. The method of claim 15, further comprising: inserting a first proximal bone fixation element through a first proximal hole and a second proximal bone fixation element through a second proximal hole into the proximal tibia segment, the first and second proximal holes extending through the proximal portion of the bone plate so that, when the bone plate is in the desired initial position, the first and second proximal bone fixation elements fix the proximal portion of the bone plate relative to the proximal tibial segment prior to insertion of the second distal bone fixation element through the second distal hole.
19. The method of claim 18, wherein the first proximal bone fixation element is inserted into a cranial side of the proximal tibial segment.